Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/10/2024
PI Name:	Goel, Namni Ph.D.		
Project Title:	Biomarkers as Predictors of Resiliency and Susceptibility to Stress in Space Flight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: Formerly at the University of Pennsylvania until July 2019.		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health (FLAGSHIP & NSBRI)
Start Date:	10/23/2019	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2025 per NASA-JSC (Ed., 9/17/24) NOTE: End date changed to 9/30/2024 per NASA-JSC (Ed., 5/2/24) NOTE: End date changed to 9/20/2024 per NSSC information (Ed., 9/3/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC20K0243		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Biomarkers as Predictors of Resiliency and Susceptibility to Stress in Space Flight," grant NNX14AN49G, due to Principal Investigator (PI) move to Rush University from University of Pennsylvania in summer 2019, requiring issue of new grant. This proposal is responsive to the NASA Behavioral Health and Performance gap (BMed5) to find individual characteristics that predict successful adaptation and performance in an isolated, confined, and extreme environment, especially for long duration missions. The project also relates to Human Research Program (HRP) Sleep Gap 4 to identify indicators of individual susceptibilities and resiliencies to sleep loss and circadian rhythm disruption, to aid with individualized countermeasure regimens, for autonomous, long duration, and/or distance exploration missions. The proposal is also responsive to BMed 1 and BMed 2, and Sleep Gap 2 and Sleep Gap 9 [Ed. note: Gap names have changed since this 2014 proposal. See https:// To address these gaps, this proposal will assess biomarkers as predictors of resiliency and susceptibility (individual differences) to performance stress and sleep loss using the HRP Human Exploration Research Analog (HERA) and the Hawaii Space Exploration Analog and Simulation (HI-SEAS) high fidelity space analog facilities. We will conduct a ground-based experiment—strongly anchored in our previous laboratory-based research—on N=32 healthy men and women (ages 26-55) in the HERA facility (short-duration analog) and on N=6 healthy men and women (ages 21-65) in the HI-SEAS facility (long-duration analog) to determine the predictive validity of a set of relevant, valid, and reliable biomarkers for distinguishing those who are more resilient versus those who are more susceptible to the adverse neurobehavioral effects of the combination of high performance demands and total sleep deprivation (TSD) stressors—two conditions commonly experienced in space flight. These biomarkers include the following: cardiovascular measures (blood pressure, heart rate and heart rate variability, stroke volume, and cardiac output), salivary cortisol, catecholamines (dopamine, noradrenaline, and adrenaline), an inflammatory marker (C-reactive protein; CRP), metabolomic markers (via unbiased metabolomics), and microRNAs (epigenetic markers). The project deliverable will be a countermeasure (set of diverse biomarkers) for distinguishing those who are more resilient versus those who are more susceptible to the adverse neurobehavioral effects of high-performance demands and sleep loss stressors. If valid markers of such susceptibility can be found, it will be possible to optimize and individualize crew resources and mitigate stress and other behavioral health and performance risks autonomously during long-duration space flight. The SIRIUS (Scientific International Research In a Unique terrestrial Station) missions are the first time NASA's Human Research Program (HRP) partners with Russia's IBMP (Institute for Biomedical Problems) Ground-based Experimental Complex, NEK (Nezemnyy Eksperimental'nyy Kompleks) to conduct a series of analog missions. Dr. Goel's project will be part of the 2019 mission as well as the 2021 mission.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project's research will deliver a countermeasure (set of diverse biomarkers) for distinguishing those who are more resilient versus those who are more susceptible to the adverse neurobehavioral effects of high performance demands and sleep loss stressors. If valid markers of such susceptibility can be found, it will be possible to optimize and individualize crew resources and mitigate stress and other behavioral health and performance risks autonomously during long-duration space flight. This information would also be of use on Earth in applied occupations that demand similar risks and stressors.
Task Progress:	We integrated the complex, multifaceted five-day stress and sleep loss experiment into Human Exploration Research Analog (HERA) and successfully collected data in all four 14-day 2015 and all four 30-day 2016 missions (N=32 crewmembers). These data include the following biomarkers: blood markers from 6 time points in 32 crewmembers (190 blood markers; n=2 crewmembers did not participate in one biomarker assessment); 2 saliva markers each from 6 time points in 32 crewmembers (382 saliva markers; n=1 crewmember did not participate in one biomarker assessment); blood pressure markers from 6 time points in 32 crewmembers (191 blood pressure markers; n=1 crewmember did not participate in one biomarker assessment); stroke volume and cardiac output from 6 time points in 32 crewmembers (191 stroke volume and cardiac output markers; n=1 crewmember did not participate in one biomarker assessment); and heart rate from 6 time points in 32 crewmembers (189 heart rate markers: 3 heart rate monitor data points were not collected due to n=2 crewmembers mistakenly not turning on the heart rate device and n=1 crewmember not participating in one biomarker assessment; however, heart rate data collected from the echocardiography and/or blood pressure devices can be used as needed). We also have data from 11 neurobehavioral tests for 32 crewmembers (348 neurobehavioral tests; one crewmember did not participate in 4 neurobehavioral assessments). Almost all the missing data can be attributed to one crewmember who experienced a medical emergency. Finally, we have continuous actigraphy data on n=16 crewmembers for 14-days each (a total of 224 days of actigraphy) and on n=16 crewmembers for 30-days each (a total of 480 days of actigraphy). Analyses of the wrist actigraphy data from the four 14-day HERA missions of 2015 (n=16) and the four 30-day HERA missions of 2016 (n=16) indicate crewmembers were compliant with the dictated sleep-wake times at baseline and recovery and were not sleeping during the total sleep deprivation (TSD) night. As expected for these 32 crewmembers, on average, the performance variables show significant impairment with TSD (with individual differences in neurobehavioral responses). Thus, the sleep loss manipulation in HERA was highly effective. We successfully completed the 17-day initial "shakedown" mission in November 2017 on N=6 subjects. Two miRNA samples were not collected due to blood flow issues with the blood draws, and one NTB test bout was not collected; all other pilot data were successfully collected. We successfully completed the 4-month, long-duration mission in NEK in July 2019 on N=6 subjects, with all data collected. We successfully completed the 8-month, long-duration mission in Nezemnyy Eksperimental'nyy Kompleks (NEK) in July 2022 on N=5 subjects, with all data collected (one subject withdrew during the mission). Unfortunately, we will not be participating in the 12-month, long-duration mission in NEK as originally planned.
Bibliography Type:	Description: (Last Updated: 02/16/2026)
Abstracts for Journals and Proceedings	Wahba M, Moore M, Goel N, Allison KC. "Adherence to a meal plan on a daytime vs delayed schedule." Penn Undergraduate Research Mentoring Program Conference 2023, Philadelphia, Pennsylvania, September 18, 2023. Abstracts. Penn Undergraduate Research Mentoring Program Conference 2023, Philadelphia, Pennsylvania, September 18, 2023. , Sep-2023

Abstracts for Journals and Proceedings	Goel N, Pasetes LN. "Biomarkers as predictors of resiliency and susceptibility to stress in space flight." 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Pasetes LN, Goel N. "Phenotypic stability of actigraphic sleep measures across long-duration intervals involving repeated sleep loss and recovery." 2024 NASA Human Research Program Investigators' Workshop 2024, Galveston, Texas, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop 2024, Galveston, Texas, February 13-16, 2024. , Feb-2024
Abstracts for Journals and Proceedings	Malone SK, Patterson F, Goyal C, Hu J, Perez N, Dickson VV, Melkus GD, Goel N, Aouizerat B. "Predicting dim light melatonin onset from a single blood sample correlates with actigraphy-estimated sleep in middle-aged adults with short sleep duration." Society for Research on Biological Rhythms Meeting, San Juan, Puerto Rico, May 18-22, 2024. Abstracts. Society for Research on Biological Rhythms Meeting, San Juan, Puerto Rico, May 18-22, 2024. , May-2024
Abstracts for Journals and Proceedings	Goel N, Pasetes LN. "Predictive relationships between baseline and recovery sleep and neurobehavioral measures during sleep deprivation." Associated Professional Sleep Societies Meeting, Houston, Texas, June 1-5, 2024. Abstracts. Associated Professional Sleep Societies Meeting, Houston, Texas, June 1-5, 2024. , Jun-2024
Abstracts for Journals and Proceedings	Pasetes LN, Rosendahl-Garcia KM, Goel N. "Baseline actigraphic sleep measures predict cardiovascular responses to sleep deprivation and psychological stress." Associated Professional Sleep Societies Meeting, Houston, Texas, June 1-5, 2024. Abstracts. Associated Professional Sleep Societies Meeting, Houston, Texas, June 1-5, 2024. , Jun-2024
Abstracts for Journals and Proceedings	Pasetes LN, Rosendahl-Garcia KM, Goel N. "Cardiovascular responses to sleep deprivation and psychological stress predict actigraphic sleep during recovery." Associated Professional Sleep Societies Meeting, Houston, Texas, June 1-5, 2024. Abstracts. Associated Professional Sleep Societies Meeting, Houston, Texas, June 1-5, 2024. , Jun-2024
Abstracts for Journals and Proceedings	Chen LL, Rickels MR, Peleckis AJ, Moore MD, Goel N, Allison KC. "Effects of meal timing on insulin sensitivity, glucose effectiveness, and free fatty acid dynamics among adults with obesity." Penn Department of Medicine Research Day, Philadelphia, Pennsylvania, June 11, 2024. Abstracts. Penn Department of Medicine Research Day, Philadelphia, Pennsylvania, June 11, 2024. , Jun-2024
Abstracts for Journals and Proceedings	Clark M, Rowley JA, Goel N, Heller HC, Gurubhagavatula I, DelRosso LM, Rodriguez A, Rice-Conboy L, Martin JL. "Healthcare provider resources for OSA management." American Association of Nurse Practitioners National Conference, Nashville, Tennessee, June 25-30, 2024. Abstracts. American Association of Nurse Practitioners National Conference, Nashville, Tennessee, June 25-30, 2024. , Jun-2024
Articles in Peer-reviewed Journals	Ballard R, Parkhurst J, Julian K, Pasetes LN, Fawcett A, Li A, Goel N, Sit DK. "Light therapy for adolescent depression: A scoping review." Curr Psychiatry Rep. 2023 Sep;25(9):373-386. https://doi.org/10.1007/s11920-023-01437-5 ; PMID: 37490215 , Sep-2023
Articles in Peer-reviewed Journals	Pasetes LN, Rosendahl-Garcia KM, Goel N. "Impact of bimonthly repeated total sleep deprivation and recovery sleep on cardiovascular indices." Physiol Rep. 2023 Oct;11(20):e15841. https://doi.org/10.14814/phy2.15841 ; PMID: 37849046; PMCID: PMC10582224 , Oct-2023
Articles in Peer-reviewed Journals	Smith LT, Walsh RFL, Goel N, Alloy LB. "Social jetlag and trajectories of mood symptoms and reward responsiveness in individuals at low-risk, high-risk, and with bipolar spectrum disorders: An ecological momentary assessment study." Psychiatry Res. 2023 Nov;329:115499. https://doi.org/10.1016/j.psychres.2023.115499 ; PMID: 37774444; PMCID: PMC10841532 , Nov-2023
Articles in Peer-reviewed Journals	Walsh RFL, Klugman J, Moriarity DP, Titone MK, Ng TH, Goel N, Alloy LB. "Reward sensitivity and social rhythms during goal-striving: An ecological momentary assessment investigation of bipolar spectrum disorders." J Affect Disord. 2024 Jan 1:344:510-518. https://doi.org/10.1016/j.jad.2023.10.114 ; PMID: 37852584; PMCID: PMC10842638 , Jan-2024
Articles in Peer-reviewed Journals	Pasetes LN, Goel N. "Short-term and long-term phenotypic stability of actigraphic sleep metrics involving repeated sleep loss and recovery." J Sleep Res. 2024 Jan 29:e14149. https://doi.org/10.1111/jsr.14149 ; PMID: 38284151; PMCID: PMC11284248 , Jan-2024
Articles in Peer-reviewed Journals	Ballard R, Parkhurst JT, Gadek LK, Julian KM, Yang A, Pasetes LN, Goel N, Sit DK. "Bright light therapy for major depressive disorder in adolescent outpatients: A preliminary study." Clocks Sleep. 2024 Jan 30;6(1):56-71. https://doi.org/10.3390/clockssleep6010005 ; PMID: 38390946; PMCID: PMC10885037 , Jan-2024
Articles in Peer-reviewed Journals	Heller HC, Herzog E, Brager A, Poe G, Allada R, Scheer F, Carskadon M, de la Iglesia HO, Jang R, Montero A, Wright K, Mouraine P, Walker MP, Goel N, Hogenesch J, Van Gelder RN, Kriegsfeld L, Mah C, Colwell C, Zeitzer J, Grandner M, Jackson CL, Prichard JR, Kay SA, Paul K. "The negative effects of travel on student athletes through sleep and circadian disruption." J Biol Rhythms. 2024 Feb;39(1):5-19. https://doi.org/10.1177/07487304231207 ; PMID: 37978840; PMCID: PMC11262807 , Feb-2024
Articles in Peer-reviewed Journals	Martin JL, Rowley J, Goel N, Heller HC, Gurubhagavatula I, DelRosso L, Rodriguez A, Clark M, Rice-Conboy L, on behalf of the "Count on Sleep" workgroups. ""Count on Sleep": An OSA Awareness Project Update." J Clin Sleep Med. 2024 Feb 1;20(2):303-307. https://doi.org/10.5664/jcsm.10864 ; PMID: 37861414; PMCID: PMC10835781 , Feb-2024
Articles in Peer-reviewed Journals	Mao T, Fang Z, Chai Y, Deng Y, Rao J, Quan P, Goel N, Basner M, Guo B, Dinges DF, Liu J, Detre JA, Rao H. "Sleep deprivation attenuates neural responses to outcomes from risky decision-making." Psychophysiology. 2024 Apr;61(4):e14465. https://doi.org/10.1111/psyp.14465 ; PMID: 37905305 , Apr-2024

Articles in Peer-reviewed Journals	Chawla S, Oster H, Duffield GE, Maronde E, Guido ME, Chabot C, Dkhissi-Benyahya O, Provencio I, Goel N, Youngstedt SD, Zi-Ching Mak N, Caba M, Nikhat A, Chakrabarti S, Wang L, Davis SJ. "Reflections on several landmark advances in circadian biology." J Circadian Rhythms. 2024 Apr 1:22:1. eCollection 2024. https://doi.org/10.5334/jcr.236 ; PMID: 38617711; PMCID: PMC11011952 , Apr-2024
Articles in Peer-reviewed Journals	Chawla S, O'Neill J, Knight MI, He Y, Wang L, Maronde E, Rodríguez SG, van Ooijen G, Garbarino-Pico E, Wolf E, Dkhissi-Benyahya O, Nikhat A, Chakrabarti S, Youngstedt SD, Zi-Ching Mak N, Provencio I, Oster H, Goel N, Caba M, Oosthuizen M, Duffield GE, Chabot C, Davis SJ. "Timely questions emerging in chronobiology: The circadian clock keeps on ticking." J Circadian Rhythms. 2024 Apr 2:22:2. eCollection 2024. doi: 10.5334/jcr.237; PMID: 38617710; PMCID: PMC11011957 , Apr-2024
Awards	Goel N. "Member, Sleep Research Society Foundation Board of Directors, 2018-Present." Jun-2024
Awards	Goel N. "Member, Sleep Research Society Board of Directors, 2018-Present." Jun-2024
Awards	Goel N. "Member and Chair, Executive Committee, Sleep Research Society, 2020-Present." Jun-2024
Awards	Goel N. "Member, Board of Directors, Associated Professional Sleep Societies, 2020-Present." Jun-2024
Awards	Goel N. "Member and Chair, Nominating Committee, Sleep Research Society, 2021-Present." Jan-2024
Awards	Goel N. "Member and Chair, Committee on Committees, Sleep Research Society, 2021-Present." Apr-2024
Awards	Goel N. "Member, COV-IRT (The Coronavirus International Research Team), 2021-Present." Jun-2024
Awards	Goel N. "Member, Obstructive Sleep Apnea Awareness Program CDC Grant, Strategic Planning Workgroup, American Academy of Sleep Medicine, 2021-Present." Jun-2024
Awards	Goel N. "Member, Study Section, Special Emphasis Panel ZHL1 CSR-Z M2 2, National Institutes of Health, 2024." Mar-2024
Awards	Goel N. "President, Associated Professional Sleep Societies, 2024-Present." Jun-2024
Awards	Goel N. "Immediate Past President, Sleep Research Society, 2024-Present." Jun-2024
Books/Book Chapters	Casale CE, Goel N. "Genetic markers of individual differences in response to sleep loss." Ed. In: Genetics of Sleep and Sleep Disorders. Gehrman P, Keene A, Grant S (eds.). Springer Nature Switzerland AG: Gewerbestrasse, Switzerland, Page: 393-416, 2024. https://doi.org/10.1007/978-3-031-62723-1_15 , Jul-2024